Curriculum Vitae/Resume

Udaysinh Rahulkumar Rathod

Contact Information

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Google Scholar

Research Interests

• Human pose detection & prediction; real-time fall-detection systems

• Multivariate time-series similarity search

• Geospatial healthcare metrics & LHA modeling

Education

M.S., Computer Systems Technology, Louisiana State University Shreveport (LSUS), Expected May 2026; GPA 4.0

Master's Thesis (ongoing, Fall 2025–Spring 2026): *3D Reconstruction Using Radio Frequencies for Through-Wall Depth & Pose* (Advisor: Subhajit Chakrabarty).
Focus: Multi-band RF (sub-6 GHz + mmWave) depth + coarse 3D pose; optional RGB fusion for semantics

B. Tech in Computer science and engineering, Parul Institute of Technology, Vadodara, India, 2020-2024.

Experience

Graduate Assistant

Machine Learning Lab Department of Computer Science, LSU Shreveport |Oct 2024 – Present

- Assisted with project planning, data processing, and model development in projects.
- Supported research paper writing, Literature review, poster preparation, and technical documentation across multiple collaborative projects.

Technical Skills

Programming Languages & Frameworks: Python, Dart, JavaScript, SQL, Flutter, PyTorch, TensorFlow, OpenCV, MediaPipe, Firebase, Langchin

Data Science & Machine Learning: LSTM, Siamese Networks, Soft-DTW, Time-Series Analysis, Supervised Learning, Unsupervised Clustering,LLM

Data Handling & Visualization: Pandas, NumPy, GeoPandas, Matplotlib, Tableau, Jupyter Notebooks

Geospatial & Remote Sensing: MODIS HDF, OSRM, Spatial Routing, Remote Sensing APIs, Geospatial Clustering,GEE

Dev Tools & Deployment: Git, Docker, Firebase, VS Code, SQL Server, REST APIs

RESEARCH & TECHNICAL PROJECTS

1. Time-Series Pattern Matching with Hybrid Siamese Networks

Collaboration with Benteler Steel & Tube / LSU Shreveport | 2024 – Present

- Developed a hybrid Soft-DTW Siamese neural network designed to learn similarity patterns in multivariate time-series data.
- Focused on replacing direct distance matrix computation with learned matching through twin neural encoding.
- Applied model to industrial manufacturing sequences and anomaly detection.
- **Outcome:** Paper currently in preparation for submission.
- **Skills:** Python, PyTorch, Siamese Networks, Soft-DTW, Time-Series Analytics, Industrial Sensor Data

2. Locational Health Access (LHA) Modeling for Cancer & Medicare Access

- Created a metric for Locational Health Access (LHA) focused on colorectal cancer access in Louisiana.
- Expanded the project nationwide using Census and Medicare datasets, introducing two novel algorithms:
  - *Water-Centroid Exclusion* for geographic filtering
  - *Hospital Radius Optimization* to reduce computational load using OSRM routing
- **Outcomes:**
  - "A New Metric for Measuring LHA," *IEEE BIBM 2024*
  - "Underserved Areas of Medicare Hospitals," *IEEE IETC 2025*
  - Posters at *AIM-AHEAD 2025* on LHA .
- **Skills:** Python, Pandas, SQL Server, OSRM, Geospatial Analysis

3. Fall Detection and Pose Monitoring

Collaboration with Neurosurgery department LSUH Shreveport / LSU Shreveport | 2024 – Present

- Developed a real-time fall detection system using MediaPipe and LSTM-based time series prediction.
- Integrated the system with clinical workflow in collaboration with LSU Health Neurology Department.
- Extended into surgical skill analysis using pose estimation from surgical videos.
- **Outcomes:**
  - "Predictive Pose Monitoring for Patient Safety," accepted, *SBEC 2025* (First Author)
  - "Kinematic Analysis of Lumbar Pedicle Screw Placement," *Neurosurgical Focus*, 2025
  - Posters at *LSUS Scholar Forum* and *Fall Prevention Symposium*, 2025
- **Skills:** MediaPipe, TensorFlow, OpenCV, Python, Jupyter, Clinical Collaboration

4. PM 2.5 Prediction from NASA MODIS Satellite Data

- Built a spatiotemporal pipeline to model PM 2.5 using surface reflectance and aerosol density from MODIS.
- Performed seasonal pattern analysis and matched predictions with AQS ground-truth data.

- **Outcome:** "Direct Prediction of PM 2.5 ," *IEEE IETC 2025*
- **Skills:** MODIS HDF, GeoPandas, NumPy, Python, Remote Sensing APIs

5. Intern – Benteler Steel & Tube (Summer & Fall 2025)

- Designed and implemented a benteler authentication service system (BASS) integrating .NET Core, SQL Server, and role-based access control.
- Developed a multimodal RAG system with custom ranking and relevance-based chunking for offline deployment.
- Leveraged CLIP and MpNet embeddings with LLaMA 3.2 LLM to enable multilingual and multimodal query answering without API dependencies.
- Enhanced scalability, security, and privacy in real-time AI deployment environments.

6. 3D Reconstruction Using Radio Frequencies

- Multi-band RF (sub-6 GHz + mmWave) through-wall depth & coarse 3D pose; physics-guided, learning-aided inverse; optional RGB fusion.
- Deliverables: benchmark protocol, code release, first-author submission (target Summer 2026)

Publications

[1] "A New Metric for Measuring Locational Health Access for Cancer Treatment," in *2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, Dec. 2024, pp. 6582–6588. doi: 10.1109/BIBM62325.2024.10822220.

[2] "A New Data Driven Approach for Identifying Underserved Areas of Hospitals Accepting Medicare," in *2025 Intermountain Engineering, Technology and Computing (IETC)*, May 2025, pp. 1–6. doi: 10.1109/IETC64455.2025.11039493.

[3] "Kinematic analysis of lumbar pedicle screw placement using an artificial intelligence framework," July 2025, doi: 10.3171/2025.4.FOCUS25157.

[4] "Direct Prediction of PM 2.5 from Raw Satellite Data – A Prospectus," in *2025 Intermountain Engineering, Technology and Computing (IETC)*, May 2025, pp. 1–6. doi: 10.1109/IETC64455.2025.11039472.

[5] "An Algorithm for Feature Based Segmentation of Contiguous Area in a Community to Locate Underserved Areas." *Proceedings of the 1st ACM SIGSPATIAL International Workshop on Spatial Intelligence for Smart and Connected Communities*. 2025.

Under review / accepted

1) REAL-TIME PATIENT SAFETY SYSTEM USING DEEP LEARNING (Journal of Biomedical... Sci-accepted)
2) PM 2.5 Estimation for US Region Using MODIS Satellite-Derived TOAR and Meteorological Variables (BIBM2025 / accepted)

Scholarships

- J. P. Morgan Chase Endowed Scholarship 2025

CURRICULAR & EXTRACURRICULAR ACTIVITIES

Theatre

- Won 1st place (Gold Medal) as a writer and actor in intercollegiate drama competition.
- Volunteered for public educational plays aimed at community awareness and literacy.